

Coding challenge overview

This is a problem that we have designed to be as similar as possible to some part of the actual daily work. We encourage you to use any tool or vibe-coding app that you are comfortable with in order to approach it. As a matter of fact, we have already implemented it completely by vibe-coding. This task is simply too large to be done by fully manual coding. You have 24 hours to complete it, with vibe-coding it should be doable in significantly less time.

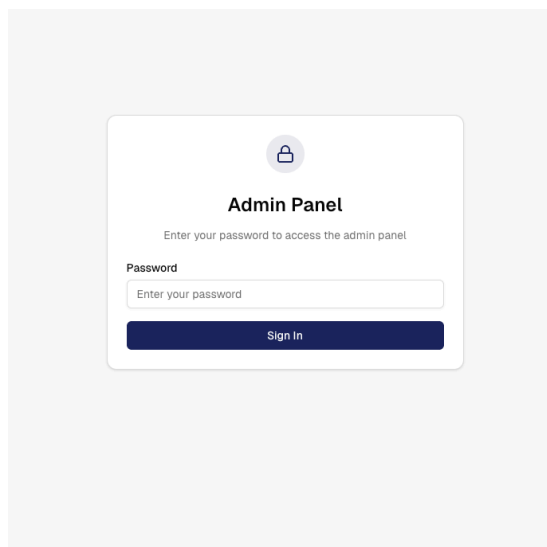
We do not expect a perfect implementation from you, but we do expect that you can reason about the requirements, understand the architecture (even if AI writes the code) and are able to articulate it well, and explain your design decisions. This is as much about evaluating communication skills as it is about evaluating technical ones.

Task

We manage a lot of data about companies and their attributes such as names, urls, headquarter countries, and sector assignment. We would like to create an Admin Panel to be used internally where non-technical colleagues can easily browse and edit parts of our database.

Each company has a few attributes (name, url, HQ country, sector(s)).
Each sector has a name and a radius attribute, and sectors can have other sectors as parents.

Below are some screenshots:



Cloud IB

Admin Panel

Companies

Sectors

Reporting

Companies

Manage your company directory

Search companies by name, CIB ID, or sector...

Name	CIB ID	Country	Sectors	Actions
바이오클로브	mally.so	KR	varies from Je	
차바이오클로브	news.chabio.com	KR	varies from Je	
한국제약바이오협회			varies from Je	
-			Healthcare Provic... neral Physicia	
051 Capital	example.com		anagement / I	
056 664 02 24			Healthcare Provic... neral Physicia	
089PHARM			varies from Je	
100XBIO			varies from Je	
103 Advisory Group			anagement / I	
1040 Tax Solutions			essional Serv... Tax Advisors	
1073849 Ontario	tollos.com	US	quipment and	
10 East Partners	10east.co	US	anagement / I	
10er Friseur	10er-friseur.de	DE	er Products &	
10KATE	10kate-laboratories.com	NL	varies from Je	

Add New Company

Enter the company details below to add them to your directory.

CIB ID

example.com

Company Name

Enter company name

HQ Country (Optional)

US, GB, DE, etc.

Cancel

Add Company

Cloud IB

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Manage your company directory

deutsche ba

Name	CIB ID	Country	Sectors	Actions
Deutsche Bank			Banks	
DEUTSCHE BANK AKTIENGESELLSCHAFT			Banks	
Deutsche Bank - Wealth Management			ther Technolo	
Flinkster (Deutsche Bahn)			essional Serv... Rental / Leas +1	

Showing 1 to 4 of 4 companies

Edit Company

Update the company details and select sectors.

CIB ID

flinkster.de

Company Name

Flinkster (Deutsche Bahn)

HQ Country (Optional)

DE

Sectors

Search sectors...

☐ Full-time Management (10 companies)

☒ Professional Services (17165 companies)

☐ Accounting Services (1828 companies)

☐ Aviation Maintenance and Repair (55 companies)

☒ Car Rental / Leasing (24 companies)

☒ Car Sharing (24 companies)

☐ ER&D (48 companies)

☐ Information Services (14 companies)

☐ Tax Advisors (11258 companies)

☐ Publishing (0 companies)

☐ Pulp and Paper (0 companies)

☐ Real Estate Investment (61 companies)

☐ Real Estate Management (10 companies)

Cancel

Save Changes

Select Companies

Choose companies to report on by uploading a CSV file or selecting sectors

Upload CSV

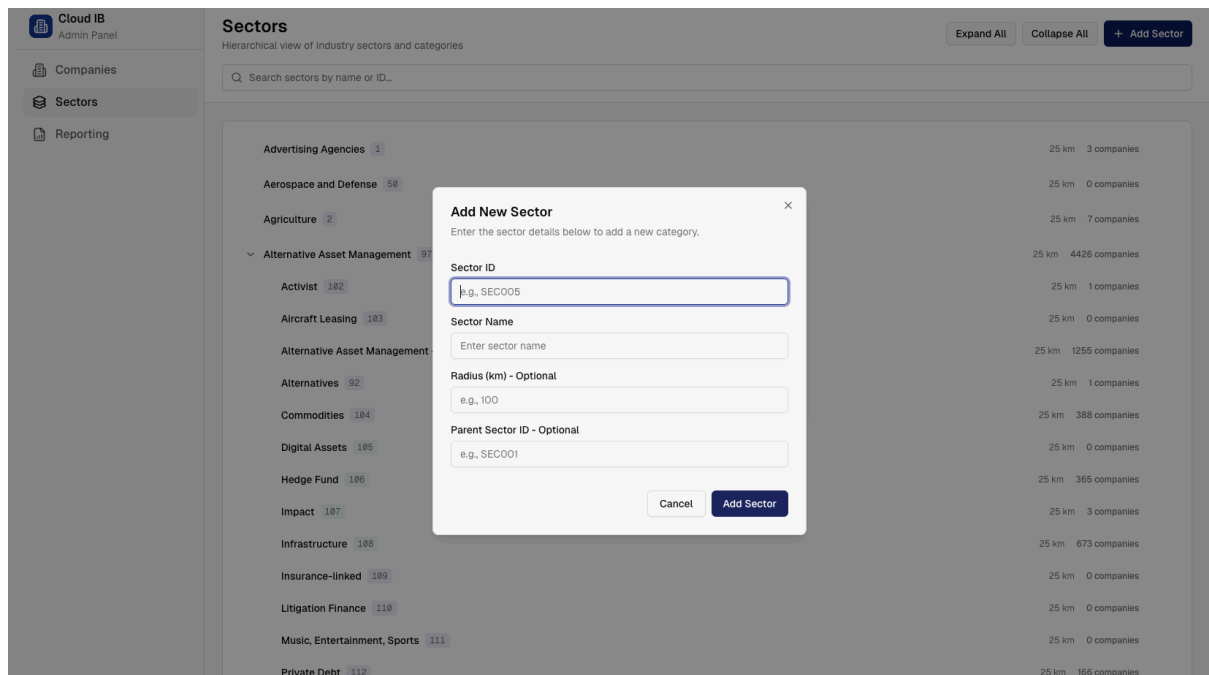
Select Sectors

Drop your CSV file here, or click to browse

File must contain either 'cib_id' or 'url' column

Select File

(on the above screen the user can upload a CSV file and get a table if/what sector each company is assigned to)



We ask you to implement a similar Admin Panel which allows for viewing / searching / linking companies and sectors. Do use the screenshots for inspiration or even follow them exactly for your UI. In a separate attachment we will send you 3 CSV files with all sectors and a sample of 1000 companies (and the linking). Feel free to augment the dataset with more synthetically-generated companies if you like.

You should create a backend in Python (using Flask / FastAPI), a front-end in React, a database (using Postgres or SQLite), and have these all containerized and ready for deployment (i.e. corresponding docker-compose, Dockerfile). If you can deploy this somewhere online and send as a link even better, but this is not a requirement. You should send us all components in a zip file (or link to a GitHub repo) and be prepared to demo the running system in a video call.

It would be particularly nice if you can explain to us where AI does well, where it falls short, what were the challenges you encountered and how to make such a project easily maintainable beyond a one-off exercise (i.e. would <https://agents.md/> help in practice?)

Good luck!